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Abstract

Social media platforms like Twitter offer scientists the opportunity to share professional, personal and scientific information with the public. This study explores the effects of these self-disclosure types. In an online between-subjects experiment ($N = 1,458$), participants rated scientists who disclosed personal information as more likable but less competent, and scientists who disclosed professional information as more competent and engaging. Subsequently, these perceptions influenced tweet engagement and certain scientific attitudes. Social presence served as a mediator between self-disclosure and perceptions. These findings broaden our understanding of science communication on Twitter as situated within a social media platform.

Keywords: *self-disclosure, science communication, social presence, Twitter, perceptions of scientists, social media*

Behind the Lab Coat: How Scientists' Self-Disclosure on Twitter Influences Source Perceptions, Tweet Engagement and Scientific Attitudes through Social Presence

The rise of social media platforms—and their sustained popularity—have revolutionized the way we communicate and engage with each other. These digital platforms have ushered in an era of increasingly individualized and personalized media use, but so too they have allowed for the diversification of *who* can create content, as well as *what* they can create (Valkenburg & Oliver, 2019). Mirroring these shifts in our media landscape are similar changes rippling across the field of science communication. While scientists have traditionally communicated with the public through outlets such as newspapers, academic journals, and television (e.g., traditional media), social media and other digital platforms have now been positioned as additional, more interactive channels for science communication (Liang et al., 2014).

One such platform commands particular attention—Twitter. With more than 200 million active users worldwide, Twitter allows people to document the world around them in 280-character tweets, as well as to like, retweet, and comment on other users' tweets (Dixon, 2022). Twitter is also home to nearly 300,000 scientists, who use the platform to share their research, communicate with the public, and network with colleagues in their field (Costas et al., 2020; Jünger & Fährnich, 2020). Already, research has begun to look at the implications of scientists on Twitter, including (but certainly not limited to) the effects of various persuasive scientific messaging strategies (e.g., Yeo et al., 2020; Zhang and Lu, 2022) and of scientists' engagement and communication with different audiences on Twitter (e.g., Côté & Darling, 2018). Broadly, these studies focus on *intentional* science communication, that is, scientists using Twitter for the purpose of disseminating their research or strategically communicating persuasive scientific messages.

However, Twitter as a social media platform allows for much more. Twitter users often share their daily experiences, comment on current events, and offer critiques shaped by their own backgrounds as a way to reflect their identities, personalities, beliefs, and values (Murthy, 2012). These types of tweets can be considered a form of self-disclosure to, whether purposefully or not, influence impressions of the user (Marwick & Boyd, 2011; Utz, 2015). Ultimately, as Murthy (2012) notes, “Twitter has everything to do with self-presentation...[it is] an important tool to say ‘look at me’ or ‘I exist’” (Murthy, 2012, p. 1062). Scientists are not exempt from using Twitter in such a way too: a 2019 study that analyzed the public engagement of one group of scientists on Twitter found that approximately a fourth of their tweets were about personal interests (and another 23% about societal and political issues), highlighting that intentional science communication is not the only focus when scientists use Twitter generally (Jünger & Fährnrich, 2020).

Twitter users who (and even those who do not) follow scientists on Twitter, then, are inevitably exposed to these other types of engagement and content. In other words, science communication on Twitter does not exist in a vacuum, or as isolated single messages. Rather, messages may be evaluated by audiences alongside a more holistic presentation of the scientist, drawn from exposure to different types of messages posted by the scientist. So while not directly “science communication,” these other content types can still have potentially significant—albeit unintentional or indirect—implications for science communication.

In this study, we examine the effects of these “other” types of content: “professional self-disclosure” and “personal self-disclosure” (defined in the next section). Specifically, our study seeks to identify how exposure to these self-disclosures influences how audiences perceive the scientist as well as downstream outcomes. Altogether, this study integrates theoretical insights from extant literature in political, science and computer-mediated communication, and media

effects broadly to offer a more comprehensive outlook towards science communication on Twitter, contextualized by the fact that Twitter is, first and foremost, a social media platform.

Scientists' Self-Disclosure on Social Media

While many definitions of self-disclosure have been proposed, the concept is generally defined as the act of revealing information about oneself to others (Derlega et al., 2008). Self-disclosure often features prominently on social media platforms, and can occur in a number of forms and manners. Research from Social Penetration Theory, which argues that self-disclosure is fundamental to the development of interpersonal relationships, identifies two main dimensions: breadth and depth (Altman & Taylor, 1973; Carpenter & Greene, 2015). The former is defined as the number and variety of topics disclosed, while the latter is defined as how much an individual chooses to disclose (otherwise called the degree of intimacy) (Carpenter & Greene, 2015; J. Kim & Song, 2016). The content itself can also vary: descriptive disclosures focus on facts about an individual, while evaluative disclosures reveal an individual's opinions and feelings.

Recent research in CMC has examined the effects of self-disclosure by different sources. Largely, such work has focused on celebrities (J. Kim & Song, 2016), influencers (Leite & Baptista, 2021; Su et al., 2021), or between users more broadly (e.g., Utz, 2015). These studies have sought to understand how self-disclosure influences key interpersonal relationship variables and attitudes towards the individual (or group) engaging in self-disclosure, as well as other behavioral intentions. Here, we argue that self-disclosure and its effects should be examined in another group: scientists on Twitter.

Scientists on Twitter are engaging in self-disclosure (and quite frequently so). Recent studies have shown that scientists often use the same Twitter account for tweeting both professional and personal information, and their personal content features prominently on their

profiles (Bowman, 2015; Ke et al., 2017). A quick scroll through the profiles of popular scientists, such as @ravenscimaven (a microbiologist with 132k followers) or @starstrickenSF (an astrophysicist with 135k followers), similarly reveals that scientists are sharing personal information about their lives and interests alongside their research, careers and other scientific messages.

Here, we focus on two types of content-oriented self-disclosure that scientists already frequently engage in on Twitter: professional self-disclosure (PFSD) and personal self-disclosure (PSD), drawn from Kim and Song's (2016) work on celebrity self-disclosure. We define PFSD as the sharing of professional experiences and research related to a scientist's career. For example, a scientist could tweet about their research or recent publications with their followers. Notably, this type of disclosure still centers the scientist; messages that are purely informative, or focus on other entities, do not qualify as PFSDs under this conceptualization. Next, we define PSD as the sharing of personal interests, hobbies and other non-science related information. This type of disclosure excludes direct references to the scientist's research in order to isolate the effects of a scientist sharing their personal life and information, outside of science, on Twitter.

In particular, we are interested in how PFSD and PSD influence outcomes related to perceptions of the scientist, which can be linked to relevant downstream outcome variables, of which we examine tweet engagement and support for scientific funding. Next, we outline current literature that has explored the potential effects of self-disclosure, particularly those of PFSD and PSD, on four source perceptions, as well as other related research.

Effects of Self-Disclosure on Source Perceptions

How people perceive scientists has become increasingly important in today's polarized, politicized and fragmented information environment. That is, audiences exposed to scientific

messages not only evaluate the message itself, but also consider various aspects of the communicator's (i.e., the scientist's) character (Fiske & Dupree, 2014). Here, we draw from the Stereotype Content Model (SCM) to identify relevant characteristics that individuals may use in their judgements towards scientists (and subsequently their messages) (Fiske et al., 2002).

SCM posits two dimensions of social perception: warmth and competence. The former includes a number of traits like trustworthiness or likability, while the latter comprises traits like intelligence and knowledge (Cuddy et al., 2008). Since we are using SCM broadly as an orienting framework, we select one trait indicative of the "competence" dimension and two on the "warmth" dimension due to their relevance in science communication research. For the former, we chose "competence" as an indicator of perceived capability. For the latter, we chose the traits "likability" and "trustworthiness," as research has shown that the two may be influenced by different factors and have unique downstream effects for science communication (Critchley, 2008; Yeo et al., 2021). Finally, we select "engagingness" to understand if self-disclosure can influence audiences' interest in the scientist more broadly.

Competence

To examine the effects of self-disclosure on perceived competence, we draw from a related concept in the realm of political communication: personalization. Personalization emphasizes a candidate-centered approach to politics, although the literature in that area has not specifically distinguished between PFSD and PSD (E.-J. Lee & Oh, 2012).

Research has shown that when candidates self-disclose information related to their campaign or the political experience (i.e., a form of PFSD), voters may see them as more competent in that particular issue (Coffé & Theiss-Morse, 2016). Specifically, when political candidates reveal information related to their occupational background, such information can help

voters determine the candidate's skills, knowledge, and competency in various policy issues (Coffé & Theiss-Morse, 2016). Thus, we may expect that scientists who share professional information about their scientific achievements and career similarly indicate to audiences their competency and capability in the realm of science.

Other studies on personalization have focused explicitly on the inclusion of personal experiences or the connection between a candidate's private life to the content within their tweets (E.-J. Lee & Oh, 2012; Meeks, 2017). The results from these studies broadly showed that personalization contributes to increased perceptions of issue competency, although factors like an individual's interpersonal dispositions and political identity moderated these results. However, we should note that under this conceptualization of "personalization," the "personal" is not entirely removed from the political, whereas our conceptualization of PSD focuses purely on the disclosure of personal information without references to science. Still, we may expect similar results, given that these personal tweets by scientists are still embedded within their overall Twitter profile. As such, we hypothesize:

H1: The presence (vs. absence) of (a) PFSD and (b) PSD will lead to an increase in perceptions of the scientist as competent.

Likability

The relationship between self-disclosure and liking has been the focus of numerous empirical studies across communication and psychology. Generally, the literature points towards the "disclosure-liking hypothesis," which has argued that higher levels of self-disclosure can lead to stronger liking effects (Collins & Miller, 1994). In their meta-analysis of nearly 100 studies that examined the relationship between self-disclosure and liking, Collins and Miller (1994) identified two general models that have been put forth to explain this relationship. The first, drawn from

Social Penetration Theory, suggests that self-disclosure is a rewarding outcome for the recipient, since it indicates that the discloser wants to further their relationship (Altman & Taylor, 1973; Collins & Miller, 1994). Thus, according to this perspective, as individuals learn more about and feel closer to a scientist through the latter's disclosure of personal information, it follows they would also like the scientist more. The second model, derived from an information-processing perspective, suggests that levels of liking are determined by the amount of information an individual holds about another. If such information is positive, the individual will like the discloser, and vice versa (Collins & Miller, 1994). Thus, under this model, a scientist who discloses professional information about their careers or scientific accomplishments (i.e., engaging in PFSD) is sharing more positive information about themselves, leading to increased liking towards them.

While Collins & Miller's meta-analysis focused solely on face-to-face interactions, their observation that people like others who disclose to them has been broadly corroborated in the CMC literature. These studies have also provided additional empirical support for both the information-processing perspective and the Social Penetration Theory (Kashian et al., 2017), and have even found that these relationships are intensified in a computer-mediated environment (Jiang et al., 2011). Therefore, we propose:

H2: The presence (vs. absence) of (a) PFSD and (b) PSD will lead to an increase in perceptions of the scientist as likable.

Trustworthiness

An emerging body of research has focused on how people evaluate various sources' trustworthiness through their online profiles (i.e., where they can self-disclose different aspects of themselves). For example, a 2017 study examined the effects of self-disclosure strategies used by Airbnb hosts on perceived trustworthiness (Ma et al., 2017). Their study found that as hosts self-

disclosed more information, the uncertainty towards their profiles decreased among profile viewers, which subsequently increased their perceived trustworthiness. More specifically, profiles that disclosed more information—and more diverse information, at that—were perceived as more trustworthy (Ma et al., 2017). Another study, focusing on teachers' self-disclosure on Facebook, found similar results (Mazer et al., 2009). Teachers who shared more personal information and opinions on relevant topics were perceived as more similar to students, which subsequently increased students' perceptions of the teacher's credibility. These results are further demonstrated in the realm of influencer marketing: influencers who engage in intimate self-disclosure can foster parasocial relationships with their followers, who then trust the influencer's opinions and advice more (Leite & Baptista, 2021).

Broadly, these results suggest that as self-disclosure increases, so do perceptions of the sources' trustworthiness. So, we would expect that as scientists share a broader range of information about themselves (from the personal to the professional), audiences exposed to these messages may then rate these scientists as more trustworthy than if they do not engage in any self-disclosure.

H3: The presence (vs. absence) of (a) PFSD and (b) PSD will lead to an increase in perceptions of the scientist as trustworthy.

Engagingness

Research that has examined specifically the impacts of self-disclosure on the perceived engagingness of a source remains sparse; thus, we turn towards extant research on personal narratives and engagement. This line of research draws from Busselle and Bilandzic's (2009) work on narrative engagement, which sought to both identify potential mechanisms that influenced such engagement and distinguish among different aspects of engagement (Busselle & Bilandzic, 2009).

While the term “engagement” here refers more broadly to how an individual engages with, or experiences, a narrative itself, it is also a useful proxy for understanding, in the context of our study, the perceived engagingness of (or the attention paid towards) the source of a narrative (or in this case, a series of tweets).

Among these mechanisms that can increase or influence engagement is perspective-taking (i.e., when an individual sees an event from the perspective of a character) (Busselle & Bilandzic, 2009). Importantly, the relationship between perspective-taking and self-disclosure has been documented in past research: as individuals self-disclose information, others can share in the experiences, or take the perspectives, of the discloser (Weber & Carter, 1998). Thus, those exposed to self-disclosure may subsequently engage in this perspective-taking, and therefore become more engaged with the content and perceive the source as more engaging. So, it may hold that when scientists self-disclose information (either of a professional or personal nature), audiences will also perceive them as more engaging.

H4: The presence (vs. absence) of (a) PFSD and (b) PSD will lead to an increase in perceptions of the scientist as engaging.

Finally, we may also expect that the presence of both PFSD and PSD may interact with each other. Because the two types of self-disclosure cover significantly different content and topics, their co-existence on a scientist's Twitter profile may differently influence these four source perceptions compared with just the presence of one or the other. However, to date, no work has examined any potential interaction effects between PFSD and PSD. As such, we ask:

RQ1: How will PFSD and PSD interact to influence perceptions of the scientist as (a) competent, (b) likable, (c) trustworthy and (d) engaging?

The Role of Social Presence

In 2017, a poll from the marketing firm Research America reported that 81% of Americans could not name a single living scientist (“Americans’ Attitudes about Science and Scientists,” 2017). Regardless of whether this reflects a lack of interest on the part of the public or the need for more engagement efforts on the part of the scientific community, the staggering statistic points to the fact that most Americans simply don’t know of any (living) scientists or have the opportunity to meet them.

Social media platforms like Twitter offer an opportunity to remedy this gap: while audiences may not be physically in the presence of a scientist, they can still communicate or interact with scientists on Twitter as if they were. This phenomenon broadly encapsulates the concept of social presence, generally considered a feeling of “awareness of the other person’s presence in online environments” (Short et al., 1976, p. 65).

Social presence can be influenced by a number of factors, ranging from the technological affordances of a platform to various other social factors, which include how much an individual knows about the person they are interacting with or the content of the communication (Biocca et al., 2003; K.-M. Lee & Nass, 2005). Self-disclosure, then, falls under these social factors that can foster social presence. Already, research has explored the relationships between self-disclosure, social presence, and online education (Swan & Shih, 2019) and celebrities (J. Kim & Song, 2016), finding that self-disclosure generally increases the social presence of the source through its expression of affect.

Equally important is the role that social presence plays in both our physiological and psychological responses in a computer-mediated environment (Lombard & Ditton, 1997). Higher social presence can lead to more involvement with media or intensify parasocial relationships (Lombard & Ditton, 1997). That is, because social presence can increase the feeling of a mediated

interaction as real and natural, certain experiences and responses may be intensified. For example, a 2017 study on the self-disclosure strategies of celebrities found that those who engaged in PFSD and PSD increased their social presence, which subsequently increased fans' parasocial interactions with them (J. Kim & Song, 2016). Other studies have found that social presence can increase overall attitudes towards or evaluations of a source, such as their perceived trustworthiness (Bente et al., 2008; E.-J. Lee, 2013).

Based on the above literature then, we may expect social presence to mediate the relationship between PFSD and PSD on the four source perception variables. In particular, if a scientist engages in PFSD or PSD, their audience will perceive an increase in social presence, as they may see the scientist as more "real" and less "removed" from the average person. Subsequently, this increase in social presence may lead to more positive evaluations of the scientist's character.

H5: PFSD will lead to an increase in social presence, subsequently increasing perceptions of the scientist as (a) competent, (b) likable, (c) trustworthy and (d) engaging.

H6: PSD will lead to an increase in social presence, subsequently increasing perceptions of the scientist as (a) competent, (b) likable, (c) trustworthy and (d) engaging.

RQ2: How will social presence mediate the interaction between PFSD and PSD to influence perceptions of the scientist as (a) competent, (b) likable, (c) trustworthy and (d) engaging?

Persuasive Effects of Source Perceptions

In this study, we are interested in two specific downstream outcomes: tweet engagement and support for scientific funding. Since the study is situated on Twitter, we conceptualize engagement as particular behaviors unique to the platform—liking, retweeting, commenting, and

following the scientist. Tweet engagement is a useful outcome to examine because 1) social media metrics can signal to others social norms and serve as heuristics for message processing and 2) promote subsequent reach (and possibly virality) of a tweet (McAlister, 2014; Spartz et al., 2017). Our second outcome examines support for scientific funding, as one of the tweets within the scientist's profile in our study explicitly advocates for increased scientific funding. The rationale for embedding one persuasive tweet within a scientist's Twitter profile is twofold: first, we wanted to understand how this persuasive message might be evaluated not in isolation, but in the context of other tweets from the same scientist. Second, we wanted to examine how PFSD and PSD, as found within these other tweets, may directly or indirectly influence how people respond to the persuasive message itself.

Research on the persuasive effects of source perceptions have largely fallen under the dual process models of persuasion (e.g., Elaboration Likelihood Model, Petty et al., 2002). Such work has argued that attributes of a source, such as their credibility, expertise, etc., can serve as heuristics (or "mental shortcuts" that can be used to evaluate a persuasive message), particularly if an individual holds no a priori attitude about the topic and lacks the opportunity and resources to form one (Kumkale et al., 2010). Because the topics chosen for this study (geosciences and astronomy) are decidedly non-politicized and non-polarized (Nisbet et al., 2015), it is reasonable to assume most individuals do not hold any prior attitudes towards these sciences and funding for these science programs. We should note that in this study, we do not examine specifically these cognitive processes nor explicitly manipulate or measure motivation and ability as is typical in persuasion studies. Rather, we assume that under certain conditions, these source perceptions can serve as heuristics, which may then increase both tweet engagement and support for scientific funding.

Given the above review of the effects of PFSD and PSD on source perceptions, as well as the mediating role of social presence, we build upon our earlier hypotheses to suggest the following relationship:

H7: The aforementioned source perceptions will be positively associated with (a) tweet engagement and (b) support for scientific funding.

The conceptual framework for this study is shown below (**Figure 1**).

[Figure 1 here]

Methods

The present experiment employed a 2 (PFSD: present or absent) $\times$ 2 (PSD: present or absent) $\times$ 2 (context: astronomy or geosciences) between-subjects design. We recruited 1,500 U.S. participants through the online participant recruitment platform Prolific (Palan & Schitter, 2018), and retained 1,458 after excluding those who failed to answer the two attention check questions correctly. Participant demographic information, as well as descriptions of the key measures employed, are included in the Appendix A in the Supplemental Materials.

Experimental Design

A Twitter profile consisting of a scientist's profile picture, biography, username and their twelve most recent tweets were employed as the stimuli for this study. The name, username and photo of the scientist, Dr. Finley Jones, were held constant across all conditions, while the biography changed to fit the particular scientific context.

The twelve tweets were divided as follows, drawing from the experimental design by Lee and Oh (2012) and Meeks (2016): three tweets for the PFSD manipulation, three tweets for the PSD manipulation, three tweets for the scientific issue manipulation, two "filler" tweets (e.g., calls for papers, conference calls, job postings), and one persuasive scientific tweet depending on the

particular context. Prior to the main experiment, tweets were pre-tested ($N = 801$) to ensure their successful manipulation of each condition; the selected tweets were also similarly rated across a number of other characteristics, such as their understandability and engagingness, to avoid potential confounding factors. The order of the twelve tweets was randomized across all participants (refer to Appendix B and D Supplemental Materials for a sample profile and the full stimuli used, as well as the results from the pre-test, respectively).

For the first manipulation, the “present” condition included three PFSD tweets, with information about the scientist’s academic and research background, experience or career plans. In the second manipulation, the “present” condition included three PSD tweets, which referred to the scientists’ personal interests and relationships. In the “absent” conditions for both manipulations, the tweets were replaced with three additional filler tweets, which were randomly selected (with no repeats) from a pool of ten filler tweets.

The scientific issue manipulation was the final manipulation. We selected two ideologically-neutral scientific topics—geosciences and astronomy—according to Nisbet et al. (2015). The three tweets for this manipulation did not contain any self-disclosure and were purely informative tweets about the scientific topic. We opted for these topics in order to understand the effects of the two types of self-disclosure on perceptions of the scientist and attitudes towards the topic outside the influence of political ideology. More so, an analysis of nearly 300,000 scientists on Twitter showed that scientists from these two areas (under “earth sciences”) tend to be overrepresented on Twitter (Costas et al., 2020). That is, scientists in these disciplines represent the second largest faction on Twitter after the social sciences and humanities, so the general Twitter public may more likely be exposed to tweets from scientists in these areas.

Results

Induction Checks

We performed a two-way analysis of variance (ANOVA), which included the main effects of the experimental conditions (PFSD and PSD) as well as their two-way interaction term as the independent variables, to test our manipulations of these experimental conditions on perceived PFSD and perceived PSD of the stimuli. There was a significant main effect of the PFSD conditions on perceived PFSD, $F(1, 1454) = 146.76, p < .001$, partial $\eta^2 = .09$. Participants perceived the Twitter profiles with PFSD as higher in perceived PFSD ($M = 5.69, SD = 0.86$) than the profiles without PFSD ($M = 5.02, SD = 1.22$). There was also a significant main effect of the PSD conditions on perceived PSD, $F(1, 1454) = 2342.61, p < .001$, partial $\eta^2 = .62$. Participants rated the Twitter profiles with PSD as higher in perceived PSD ($M = 5.07, SD = 1.03$) than the profiles without PSD ($M = 2.43, SD = 1.05$). No interaction effects were found. As such, the ANOVAs indicated successful manipulations of the PFSD and PSD experimental conditions.

Hypotheses Tests

First, we employed a series of general linear models to test for the main effects of the experimental conditions on all of our outcome variables—perceptions of the scientist as (1) competent, (2) likable, (3) trustworthy, (4) engaging, as well as (5) social presence, (6) tweet engagement, and (7) support for scientific funding (**Table 1**). These models included the main effects of the experimental conditions (PFSD, PSD, and scientific context) as well as their two-way and three-way interaction terms as the independent variables. We found significant main effects of PFSD on perceived competence ($F(1, 1450) = 5.31, p = 0.02$), engagingness ($F(1, 1450) = 11.57, p < 0.001$), social presence ($F(1, 1450) = 5.29, p = 0.02$). Post hoc comparisons based on Bonferroni corrections showed that the presence of PFSD increased perceived competence, engagingness, and social presence. These results provide support for H1a and H4a; H2a and H3a

were not supported. There were also significant main effects of PSD on perceived competence ($F(1, 1450) = 18.68, p < 0.001$), likability ($F(1, 1450) = 10.07, p < 0.01$) and social presence ($F(1, 1450) = 6.22, p = 0.01$), such that the presence of PSD increased the likability and social presence of the scientist but decreased their perceived competence. These results provide support for H2b only; H3b and H4b were not supported, while a relationship in the opposite direction was found for H1b. Next, we found significant main effects of the scientific context on perceived engagingness ($F(1, 1450) = 6.30, p = 0.01$) and tweet engagement ($F(1, 1450) = 4.52, p = 0.03$). Specifically, the astronomy context increased both the engagingness of the scientist as well as intentions to engage with the tweet and scientist. Finally, we did not find any interaction effects between PFSD and PSD on any of the outcome variables, addressing RQ1, or any interaction effects involving the scientific context condition.

Next, we used Hayes' PROCESS 4.0 macro (2018) for IBM SPSS Statistics for our mediation analyses, with a 95% confidence interval (CI) based on 5,000 bootstrapped samples. We created three dummy variables to represent each of the three manipulations, where the reference groups were the "absent" conditions for both the PFSD and PSD manipulations (absent = -1, present = 1) and the "geosciences" condition for the scientific context manipulation (geosciences = -1, astronomy = 1).

Specifically, we used Model 4 to test H5 and H6, which predicted that social presence would mediate the relationship between the PFSD and PSD experimental manipulations and the four source perceptions. Our results showed that social presence was a significant mediator for PFSD and trustworthiness ($b_{indirect} = 0.02, SE_{indirect} = .01, 95\% CI = [0.00, 0.04]$), competence ($b_{indirect} = 0.02, SE_{indirect} = .01, 95\% CI = [0.00, 0.03]$), engagingness ($b_{indirect} = 0.03, SE_{indirect} = .01, 95\% CI = [0.00, 0.06]$) and likability ($b_{indirect} = 0.02, SE_{indirect} = .01, 95\% CI = [0.00, 0.05]$).

Similarly, social presence was also a significant mediator between PSD and trustworthiness ($b_{indirect} = 0.02$, $SE_{indirect} = .01$, 95% $CI = [0.00, 0.04]$), competence ($b_{indirect} = 0.02$, $SE_{indirect} = .01$, 95% $CI = [0.00, 0.03]$), engagingness ($b_{indirect} = 0.03$, $SE_{indirect} = .01$, 95% $CI = [0.01, 0.06]$) and likability ($b_{indirect} = 0.03$, $SE_{indirect} = .01$, 95% $CI = [0.01, 0.05]$). Thus, both the presence of PFSD and PSD (compared to their absence) increased social presence, which subsequently led to an increase in perceived trustworthiness, competence, engagingness, and likability. As such, we find support for H5 and H6. However, we again did not find any interaction effects, addressing RQ2.

Finally, we employed Model 81 to examine the full sequential mediation model proposed earlier (see Figure 1), with social presence as the first mediator, the four source perceptions as parallel subsequent mediators, and tweet engagement and support for scientific funding as the two outcomes. The full model results are presented in Figure 2.

[Figure 2 here]

We found that only certain source perceptions influenced these outcomes. Specifically, perceived engagingness of the scientist increased both tweet engagement ($b = 0.55$, $SE = 0.04$, $p < 0.01$) and support for scientific funding ($b = 0.13$, $SE = 0.05$, $p < 0.01$). Perceived trustworthiness also increased both tweet engagement ($b = 0.25$, $SE = 0.06$, $p < 0.01$) and support for scientific funding ($b = 0.37$, $SE = 0.07$, $p < 0.01$). Therefore, H7 was partially supported.

The indirect effect results for both tweet engagement and support for scientific also revealed that PFSD indirectly influenced support for scientific funding through perceived engagingness and indirectly influenced tweet engagement through both social presence and perceived engagingness (refer to Appendix C in the Supplementary Materials). In addition, PSD indirectly influenced support for scientific funding through social presence and perceived trustworthiness and indirectly influenced tweet engagement through social presence. Lastly, the

results for the sequential mediation analyses showed that both PFSD and PSD indirectly influenced both tweet engagement and support for scientific funding through social presence and then perceived trustworthiness, as well as through social presence and then perceived engagingness.

Discussion

Twitter has offered scientists an unprecedented opportunity to directly communicate and engage with audiences on scientific topics. But as much as Twitter has been touted as a medium for (scientific) information sharing, it is also a *social* media platform. Thus, Twitter has also become somewhat of a window into the “behind-the-scenes” of a scientist, allowing the public to see both the professional and the personal. What are the potential implications, then, of this more comprehensive presentation of the scientist that the public now has access to? This study sought to provide an initial examination of that question, revealing that (1) a scientist’s professional self-disclosure (PFSD) and personal self-disclosure (PSD) can influence perceptions towards them, (2) this relationship is mediated by social presence, and (3) these perceptions subsequently impact tweet engagement and specific scientific attitudes.

First, we found that both PFSD and PSD can influence certain source perceptions. The presence of PFSD on a scientist’s Twitter increased their perceived competence and engagingness, but did not influence their likability or trustworthiness. However, PSD increased their likability, decreased their competence, and did not influence perceptions of their trustworthiness or engagingness. We also did not find any interaction effects between PFSD and PSD. These results were unexpected; particularly, that PSD actually *decreased* a scientist’s perceived competence ran contrary to our initial hypotheses. So, what might explain these discrepancies?

Returning to our earlier discussion on the Stereotype Content Model, past work has shown that most social groups hold a “mixed” combination of the two dimensions; for example, scientists

and experts fall on high competence, low warmth quadrant (Fiske & Dupree, 2014). In addition, prior research has shown that the experimental manipulation of one dimension can influence the other unmanipulated dimension (Judd et al., 2005; Yzerbyt et al., 2008). Specifically, a “compensatory” process may occur if two groups are comparatively judged, while a “halo” effect can occur when only a single group is judged (Judd et al., 2005). In other words, if a group is evaluated as more positive in one dimension compared to another, the latter group will be evaluated as more positive on the second dimension. But if only one individual or group is the target, then positive evaluations in one dimension might lead to positive evaluations in the other. Since the target of our study focuses only on one individual (a scientist), we expected to see more of a “halo” effect, as outlined in the literature review. However, our results pointed to a more “compensatory” process even without the presence of an explicit comparison group, suggesting two important considerations for future research. First, audiences may hold preconceived attitudes and ideas towards scientists *as a social group* that may be difficult to change even through the behaviors of an individual scientist. Second, the results suggest a need for incorporating literature on stereotypes and social groups in science communication to more explicitly consider the implications of scientists as an “out-group” for certain segments of the audience.

Another possible explanation for these discrepancies may be found in our particular assumptions surrounding the conceptualizations of PFSD and PSD. Both assume that the nature of these disclosures is true and qualitatively good. But warranting theory (Walther & Parks, 2002) would suggest that audiences may not unanimously trust or believe this information since online self-disclosure can be easily manipulated. So, it might be plausible that a personal self-disclosure, in particular, would not be easily corroborated and therefore seen as holding less veracity, as opposed to a professional self-disclosure (e.g., a paper publication or a job promotion would hold

slightly more warranting value, since they can be ostensibly verified). Perhaps, then, PSD could elicit some form of reactance that negatively influences perceived competence, even if audiences still might see the scientist as more likable.

Next, we found that both PFSD and PSD indirectly influenced source perceptions through social presence. Both types of self-disclosure increased the social presence of the scientist; that is, scientists who shared information about themselves were seen as more “real” and “present.” Subsequently, they were more likely to be perceived as competent, likable, trustworthy and engaging. Our findings are broadly consistent with previous studies that have shown self-disclosure to be an important social factor that can increase social presence (J. Kim & Song, 2016; Swan & Shih, 2019). Furthermore, these results extend the current literature by confirming the role of social presence as a mechanism underlying the effects of self-disclosure on source perceptions.

Finally, the sequential mediation analyses revealed the impacts of these source perceptions, after they have been formed, on tweet engagement and support for scientific funding. As perceptions of the scientist as trustworthy and engaging increased, so too did tweet engagement and support. Given that these results were found even though only one of the twelve tweets shown on the scientist's profile explicitly advocated for scientific funding, they provide support for the need to consider a persuasive scientific message on Twitter in the context of the rest of a scientist's tweets. Still, that perceived competence and likability did not influence these downstream outcomes was surprising, given that these source perceptions were impacted directly by different self-disclosure types. Perhaps, then, these results also suggest that certain heuristics and perceptions (in this case, trustworthiness and engagingness) may matter more when audiences are exposed to tweets from scientists; future research could tease out potential differences among relevant source perceptions.

Altogether, this study contributes to the current literature in three key ways. First, this study emphasizes the need to consider a more holistic approach to science communication as it occurs on social media. While social media platforms like Twitter do indeed offer scientists increased exposure, and direct access, to a wider audience, their scientific messages are now impacted by their prior tweets, or at least by the impressions formed from those other tweets (which may not even be related to the strategic message itself). Second, we focus on two different types of self-disclosure (PFSD and PSD) as content within these “other tweets” that may influence perceptions of a scientist. Unlike most scientific messages found in traditional media or science blogs, Twitter offers an opportunity for scientists to center their personal and scientific identity as they communicate with potentially large and diverse audiences. Certainly, our results suggest that the implications of such are complex and nuanced; at the very least, they point to a need for researchers to carefully consider how scientists are perceived as a unique social group, how that identity may interact with a scientist’s existing identities, and how together these perceptions might inform the public’s expectations, attitudes and social norms surrounding how scientists engage with them. Lastly, our study confirms the importance of social presence as both a variable of interest to science communication scholars, as well as a potential avenue for scientists on social media to consider how they might best approach and manage their social media presence broadly. While the direct effects of PFSD and PSD were mixed, we did find that the two self-disclosure types increased the different source perceptions *through* social presence. Thus, rather than suggesting here that scientists tweet less about their personal lives or more about their accomplishments, we instead emphasize that even simply the presence of scientists on social media can have important benefits for how people perceive and interact with scientists. Of course, the ramifications of what scientists choose to *do* with their presence, then, is less straightforward; regardless, scientists should be

cognizant of this inherent tension, that audiences are responding to not only their scientific content but *all* of their content.

Study Limitations

We note several limitations in this study. First, our sample was drawn from a population of Prolific users, which limits the generalizability of this study. Second, our manipulation for the PFSD condition was not as clear as that for the PSD condition; less than a one point difference (on a 7-point Likert scale) existed between profiles that excluded and those that included PFSD; for PSD, this difference was nearly three points. While the differences for both were still significant at a 0.05 level, this might have minimized the effects of PFSD observed in this study. One potential reason for this smaller induction for PFSD (compared to PSD) may be due to our conceptualization of filler tweets. These tweets were meant to mimic job postings, call for papers, or conference calls that scientists can retweet on Twitter. However, to avoid any confounds from introducing a new technological affordance—retweeting—we opted to have the scientist themselves post the filler tweets. As such, participants might have perceived these filler tweets to be another form of PFSD. Thus, future studies could seek to disentangle these effects and examine the role of retweeting on perceptions of scientists separately. Finally, our conceptualizations of self-disclosure only focused on positively-valenced information, which may not capture the larger landscape of scientists' disclosure on Twitter. Again, this could be an area of interest for future research in order to fully understand how the breadth and depth of scientists' self-disclosure influence various attitudes towards them.

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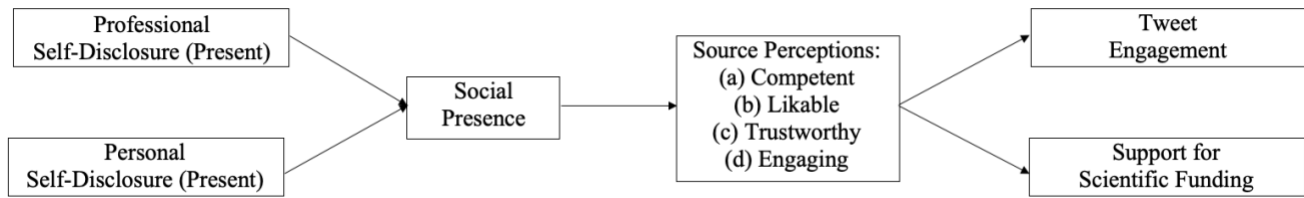


Figure 1. Conceptual framework.

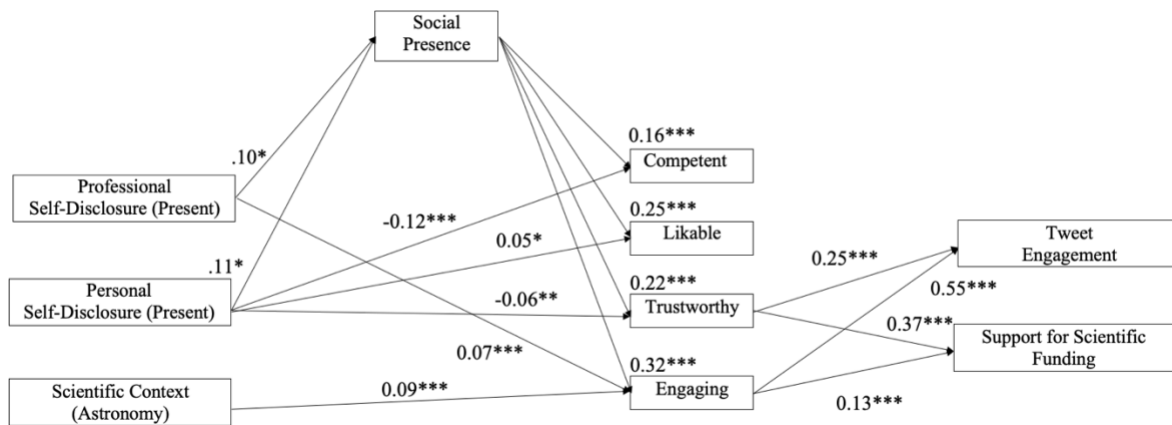


Figure 2. Full model with unstandardized coefficients. Non-significant paths are not shown, as well as the significant positive paths from social presence to both tweet engagement ($b = 0.31^{***}$) and support for scientific funding ($b = 0.05^*$). $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 1. Main effects of experimental manipulations on all variables.

	Competent	Likable	Trustworthy	Engaging	Social Presence	Tweet Engagement	Support for Funding
PFSD (Present)	0.11* (0.05)	0.05 (0.05)	0.09 ⁺ (0.05)	0.21*** (0.06)	0.20* (0.09)	0.15 ⁺ (0.08)	-0.04 (0.08)
PSD (Present)	-0.20*** (0.05)	0.16** (0.05)	-0.08 (0.05)	-0.02 (0.06)	0.21** (0.09)	0.04 (0.08)	-0.13 (0.08)
Context (Astro.)	0.05 (0.05)	0.00 (0.05)	0.01 (0.05)	0.15** (0.06)	-0.08 (0.09)	0.18* (0.08)	0.10 (0.08)
R²	0.019	0.008	0.005	0.012	0.011	0.006	0.004

Mean differences based on estimated marginal means between the present and absent conditions (for both PFSD and PSD) or between astronomy and geosciences contexts and standard errors. ⁺ $p < .10$, $^*p < .05$, $^{**}p < .01$, $^{***}p < .001$